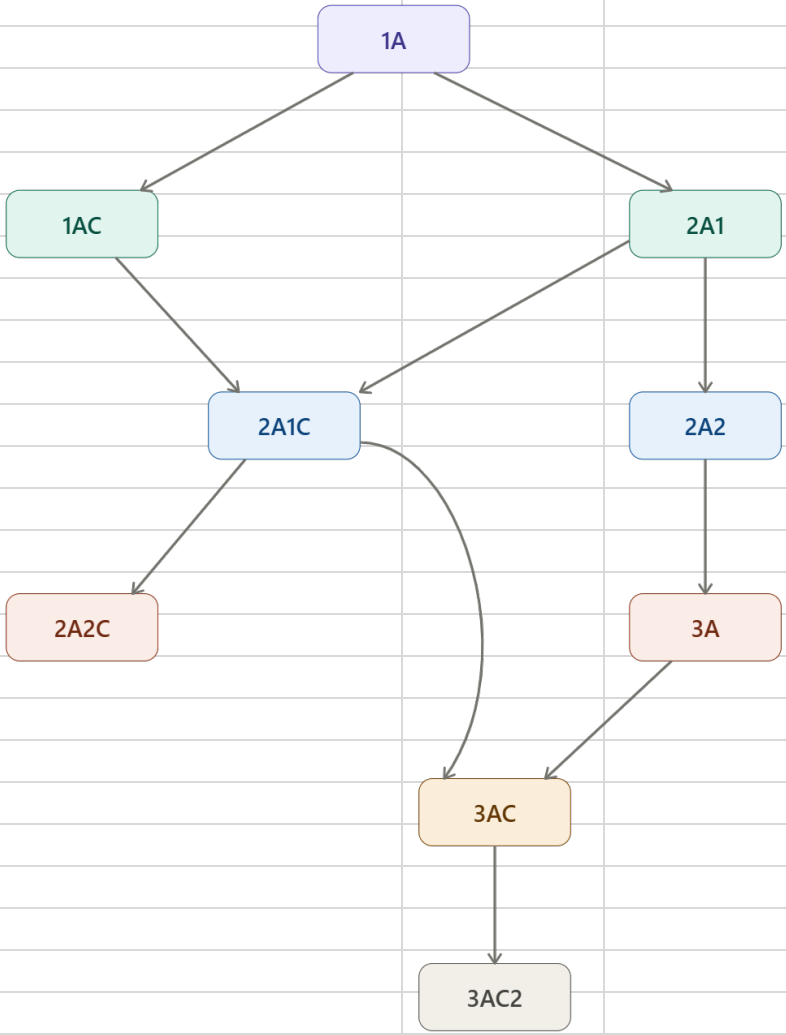
Code	Skill	Example	Misconceptions	Dependencies
1A	1-digit addition - no carry	2+2	make an off-by-one error	
1AC	1-digit addition - carry	9+4	make an off-by-one error, forget to add the carry	1A
2A1	2+1 digit addition - no carry	11+5	make an off-by-one error, add the second number at the wrong place value	1A
2A2	2+2 digit addition - no carry	11+22	make an off-by-one error, add the second number at the wrong place value	2A1
2A1C	2+1 digit addition - carry	17+9	make an off-by-one error, forget to add the carry, add the second number at the wrong place value	2A1, 1AC
2A2C	2+2 digit addition - double carry	96+89	make an off-by-one error, forget to add one of the carries, add the second number at the wrong place value	2A1C
3A	3-digit addition - no carry	123+321	make an off-by-one error, add the second number at the wrong place value	2A2
3AC	3-digit addition - single carry	369+12	make an off-by-one error, forget to add the carry, add the second number at the wrong place value	2A1C, 3A

	3AC2	3-digit addition - double carry	369+243	make an off-by-one error, forget to add one of the carries, add the second number at the wrong place value	3AC



Knowledge Boundary Detection

Step 0 — Initial scores (all 9 nodes uncolored)

Node	n^+	n^-	min
2A1C	**4**	**4**	**4** ← winner
2A2	3	4	3
3A	4	3	3
1AC	2	5	2
2A1	2	7	2
3AC	7	2	2
1A	1	9	1
2A2C	5	1	1
3AC2	8	1	1

→ ASK [2A1C]

ASK [2A1C]
│
├── ✓ CORRECT → solvable: {1A, 1AC, 2A1, 2A1C}
│ │ Remaining: {2A2, 2A2C, 3A, 3AC, 3AC2}
│ │ Scores: 3A(min=2)★ tie with 3AC(min=2) → 3A chosen

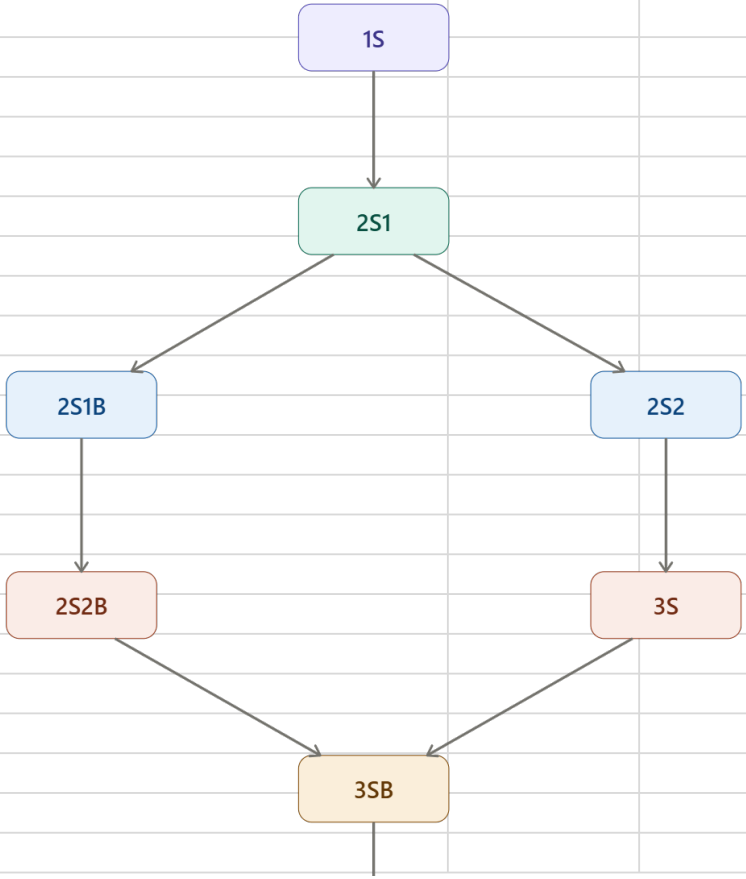
```
|  |
|  ASK [3A]
|  |
|  ├── ✓ CORRECT → solvable: {2A2, 3A}
|  |  |  Remaining: {2A2C, 3AC, 3AC2} — all min=1, tied → 2A2C chosen
|  |  |
|  |  ASK [2A2C]
|  |  |
|  |  ├── ✓ CORRECT → solvable: {2A2C}
|  |  |  |  Remaining: {3AC, 3AC2} → ASK [3AC]
|  |  |  |
|  |  |  ASK [3AC]
|  |  |  ├── ✓ CORRECT → solvable: {3AC}
|  |  |  |  ASK [3AC2]
|  |  |  |  ├── ✓ CORRECT  → ALL SOLVABLE        boundary = {}
|  |  |  |  └── ✗ WRONG    → unsolvable: {3AC2}  boundary = {3AC}
|  |  |  |
|  |  |  └── ✗ WRONG → unsolvable: {3AC, 3AC2}    boundary = {2A1C, 3A}
|  |  |
|  |  └── ✗ WRONG → unsolvable: {2A2C}
|  |     ASK [3AC]
|  |     ├── ✓ CORRECT → solvable: {3AC}
|  |     |  ASK [3AC2]
|  |     |  ├── ✓ CORRECT  boundary = {2A1C}
|  |     |  └── ✗ WRONG    boundary = {2A1C, 3AC}
|  |     |
|  |     └── ✗ WRONG → unsolvable: {3AC, 3AC2}    boundary = {2A1C, 3A}
|  |
|  └── ✗ WRONG → unsolvable: {3A, 3AC, 3AC2}
|     |  Remaining: {2A2, 2A2C} — both min=1 → 2A2 chosen
|     |
```

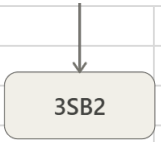
	│ ASK [2A2]	
	│ ├── ✓ CORRECT → solvable: {2A2}	
	│ │ ASK [2A2C]	
	│ │ ├── ✓ CORRECT boundary = {2A1C, 2A2}	
	│ │ └── ✗ WRONG boundary = {2A1C, 2A2}	
	│ │	
	│ └── ✗ WRONG → unsolvable: {2A2}	
	│ ASK [2A2C]	
	│ ├── ✓ CORRECT boundary = {2A1, 2A1C}	
	│ └── ✗ WRONG boundary = {2A1, 2A1C}	
	│	
	└── ✗ WRONG → unsolvable: {2A1C, 2A2C, 3AC, 3AC2}	
	│ Remaining: {1A, 1AC, 2A1, 2A2, 3A}	
	│ Scores: 2A1(min=2)★ tie with 2A2(min=2) → 2A1 chosen	
	│	
	ASK [2A1]	
	│	
	├── ✓ CORRECT → solvable: {1A, 2A1}	
	│ │ Remaining: {1AC, 2A2, 3A} — all min=1, tied → 1AC chosen	
	│ │	
	│ ASK [1AC]	
	│ │	
	│ ├── ✓ CORRECT → solvable: {1AC}	
	│ │ ASK [2A2]	
	│ │ ├── ✓ CORRECT → solvable: {2A2}	
	│ │ │ ASK [3A]	
	│ │ │ ├── ✓ CORRECT boundary = {1AC, 2A1, 3A}	
	│ │ │ └── ✗ WRONG boundary = {1AC, 2A1, 2A2}	
	│ │ │	
	│ │ └── ✗ WRONG → unsolvable: {2A2, 3A} boundary = {1AC, 2A1}	
	│ │	

	│ └── ✗ WRONG → unsolvable: {1AC}	
	│ ASK [2A2]	
	│ ├── ✓ CORRECT → solvable: {2A2}	
	│ │ ASK [3A]	
	│ │ ├── ✓ CORRECT boundary = {1A, 2A1, 3A}	
	│ │ └── ✗ WRONG boundary = {1A, 2A1, 2A2}	
	│ │	
	│ └── ✗ WRONG → unsolvable: {2A2, 3A} boundary = {1A, 2A1}	
	│	
	└── ✗ WRONG → unsolvable: {2A1, 2A2, 3A}	
	│ Remaining: {1A, 1AC} — both min=1 → 1A chosen	
	│	
	ASK [1A]	
	├── ✓ CORRECT → solvable: {1A}	
	│ ASK [1AC]	
	│ ├── ✓ CORRECT boundary = {1A, 1AC}	
	│ └── ✗ WRONG boundary = {1A}	
	│	
	└── ✗ WRONG → ALL UNSOLVABLE boundary = {1A}	

Code	Skill	Example	Misconceptions	Dependencies
1S	1-digit subtraction - no borrow	9-5	make an off-by-one error, adding instead of subtracting	
2S1	2-1 digit subtraction - no borrow	19-4	make an off-by-one error, adding instead of subtracting, subtract the second number at the wrong place value	1S
2S1B	2-1 digit subtraction - borrow	14-9	make an off-by-one error, adding instead of subtracting forget to add one of the carries, subtract the second number at the wrong place value	2S1
2S2	2-2 digit subtraction - no borrow	19-14	make an off-by-one error, adding instead of subtracting, subtract the second number at the wrong place value	2S1
3S	3-digit subtraction - no borrow	999-888	make an off-by-one error, adding instead of subtracting, subtract the second number at the wrong place value	2S2
2S2B	2-2 digit subtraction - borrow	32-19	make an off-by-one error, adding instead of subtracting forget to add one of the carries, subtract the second number at the wrong place value	2S1B

	3SB	3-digit subtraction - single borrow	991-222	make an off-by-one error, adding instead of subtracting forget to add one of the carries, subtract the second number at the wrong place value	2S1B, 3S
	3SB2	3-digit subtraction - double borrow	901-222	make an off-by-one error, adding instead of subtracting forget to add one of the carries, subtract the second number at the wrong place value	3SB





Knowledge Boundary Detection

Part (i) — Step-by-step algorithm walkthrough

Step 0 — Initial scores (all 8 nodes uncolored)

Node	n^+	n^-	min
2S1B	**3**	**4**	**3** ←
2S2	**3**	**4**	**3** ←
2S2B	**4**	**3**	**3** ←
3S	**4**	**3**	**3** ←
2S1	2	7	2
3SB	7	2	2
1S	1	8	1
3SB2	8	1	1

Four-way tie at min=3: {2S1B, 2S2, 2S2B, 3S} → **2S1B** chosen (alphabetical).

→ **ASK [2S1B]**

ASK [2S1B]
│
├── ✓ CORRECT → solvable: {1S, 2S1, 2S1B}
│ │ Remaining: {2S2, 2S2B, 3S, 3SB, 3SB2}
│ │ Scores: 2S2(min=1), 2S2B(min=1), 3S(min=2)★, 3SB(min=2)★, 3SB2(min=1)

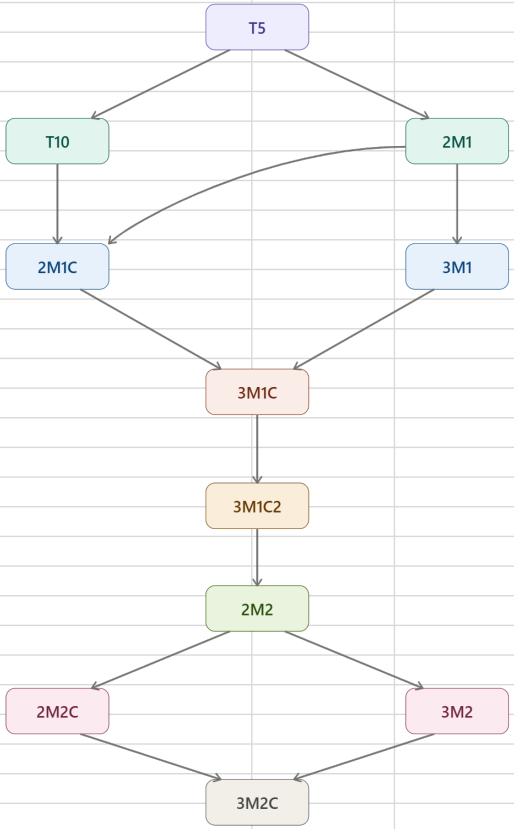
```
|  |  Tie: {3S, 3SB} → 3S (alphabetical)
|  |
|  ASK [3S]
|  |
|  ├── ✓ CORRECT → solvable: {2S2, 3S}
|  |  |  Remaining: {2S2B, 3SB, 3SB2}
|  |  |  Scores: 2S2B(min=1), 3SB(min=2)★, 3SB2(min=1)
|  |  |
|  |  ASK [3SB]
|  |  |
|  |  ├── ✓ CORRECT → solvable: {2S2B, 3SB}
|  |  |  |  Remaining: {3SB2}
|  |  |  |
|  |  |  ASK [3SB2]
|  |  |  ├── ✓ CORRECT → ALL SOLVABLE     boundary = {3SB2}
|  |  |  └── ✗ WRONG   → unsolvable: {3SB2}
|  |  |                  boundary = {3SB}
|  |  |
|  |  └── ✗ WRONG → unsolvable: {3SB, 3SB2}
|  |     |  Remaining: {2S2B}
|  |     |
|  |     ASK [2S2B]
|  |     ├── ✓ CORRECT → solvable: {2S2B}
|  |     |            boundary = {2S2B, 3S}
|  |     └── ✗ WRONG   → unsolvable: {2S2B}
|  |                  boundary = {2S1B, 3S}
|  |
|  └── ✗ WRONG → unsolvable: {3S, 3SB, 3SB2}
|     |  Remaining: {2S2, 2S2B}
|     |  Scores: 2S2(min=1)★, 2S2B(min=1)★  — tie → 2S2 (alphabetical)
|     |
|     ASK [2S2]
|     |
```

```
|       ├── ✓ CORRECT → solvable: {2S2}
|       |   |  Remaining: {2S2B}
|       |   |
|       |   ASK [2S2B]
|       |   ├── ✓ CORRECT → solvable: {2S2B}
|       |   |           boundary = {2S2, 2S2B}
|       |   └── ✗ WRONG   → unsolvable: {2S2B}
|       |               boundary = {2S1B, 2S2}
|       |
|       └── ✗ WRONG → unsolvable: {2S2}
|           |  Remaining: {2S2B}
|           |
|          ASK [2S2B]
|           ├── ✓ CORRECT → solvable: {2S2B}
|           |           boundary = {2S1, 2S2B}
|           └── ✗ WRONG   → unsolvable: {2S2B}
|                       boundary = {2S1, 2S1B}
|
└── ✗ WRONG → unsolvable: {2S1B, 2S2B, 3SB, 3SB2}
|  Remaining: {1S, 2S1, 2S2, 3S}
|  Scores: 1S(min=1), 2S1(min=2)★, 2S2(min=2)★, 3S(min=1)
|  Tie: {2S1, 2S2} → 2S1 (alphabetical)
|
ASK [2S1]
|
├── ✓ CORRECT → solvable: {1S, 2S1}
|   |  Remaining: {2S2, 3S}
|   |  Scores: 2S2(min=1)★, 3S(min=1)★  — tie → 2S2 (alphabetical)
|   |
|   ASK [2S2]
|   |
|   ├── ✓ CORRECT → solvable: {2S2}
|   |   |  Remaining: {3S}
```

	│ │ │	
	│ │ ASK [3S]	
	│ │ ├── ✓ CORRECT → solvable: {3S}	
	│ │ │ boundary = {2S1, 3S}	
	│ │ └── ✗ WRONG → unsolvable: {3S}	
	│ │ boundary = {2S1, 2S2}	
	│ │	
	│ └── ✗ WRONG → unsolvable: {2S2, 3S} boundary = {2S1}	
	│	
	└── ✗ WRONG → unsolvable: {2S1, 2S2, 3S}	
	│ Remaining: {1S}	
	│	
	ASK [1S]	
	├── ✓ CORRECT → solvable: {1S} boundary = {1S}	
	└── ✗ WRONG → ALL UNSOLVABLE boundary = {1S}	

Code	Skill	Example	Misconceptions	Dependencies
T5	Multiplication tables - up to 5	3x4	confuse one table with another	
T10	Multiplication tables - 5 to 10	9x7	confuse one table with another	T5
2M1	2x1 multiplication - no carry	13x2	make an off-by-one error, add instead of multiply	T5
3M1	3x1 multiplication - no carry	123x2	make an off-by-one error, add instead of multiply	2M1
2M1C	2x1 multiplication - carry	12x9	make an off-by-one error, add instead of multiply, forget to add one of the carries	T10
3M1C	3x1 multiplication - single carry	246x2	make an off-by-one error, add instead of multiply, forget to add one of the carries	2M1C
3M1C2	3x1 multiplication - double carry	246x4	make an off-by-one error, add instead of multiply, forget to add one of the carries, multiply the second number at the wrong place value	3M1C
2M2	2x2 multiplication - no carry (multiplication part should not have carry)	31x12	make an off-by-one error, add instead of multiply, forget to multiply the other place values, forget to add carries during addition	3M1C2
2M2C	2x2 multiplication - carry	69x69	make an off-by-one error, add instead of multiply, forget to add one of the carries, multiply the second number at the wrong place value	2M2

3M2	3x2 multiplication - no carry	123x21	make an off-by-one error, add instead of multiply, forget to multiply the other place values, forget to add carries during addition	2M2
3M2C	3x2 multiplication - carry	479x34	make an off-by-one error, add instead of multiply, forget to add one of the carries, multiply the second number at the wrong place value	3M2, 2M2C



Levels

1	T5, T10
2	2M1, 2M1C, 3M1, 3M1C, 3M1C2
3	2M2, 2M2C, 3M2, 3M2C

Code	Skill	Example	Misconceptions	Dependencies
2D1	2/1 division without remainder	81/3	make an off-by-one error, make mistakes while adding or subtracting during division	
3D1	3/1 division without remainder	123/3	make an off-by-one error, make mistakes while adding or subtracting during division	2D1
2D1R	2/1 division with remainder	70/4	make an off-by-one error, make mistakes while adding or subtracting during division	2D1
3D1R	3/1 division with remainder	123/5	make an off-by-one error, make mistakes while adding or subtracting during division	3D1, 2D1R
3D1Z	3/1 division with 0 in quotient	601/5	make an off-by-one error, make mistakes while adding or subtracting during division, forget to account for the 0 in the quotient	3D1R
4D1R	4/1 division with remainder	1001/6	make an off-by-one error, make mistakes while adding or subtracting during division	3D1R
3D2	3/2 division without remainder	207/23	make an off-by-one error, make mistakes while adding or subtracting during division	3D1
3D2R	3/2 division with reminder	829/75	make an off-by-one error, make mistakes while adding or subtracting during division	3D2, 3D1R
4D2R	4/2 division with reminder	1056/25	make an off-by-one error, make mistakes while adding or subtracting during division	3D2R, 4D1R

					Levels	
					1	2D1, 2D1R
					2	3D1, 3D1R, 3D1Z
					3	3D2, 3D2R, 4D2R, 4D1R
					0R-1D: 0 Reminder, 1-digit Divisor	